Certified Weil Positivity in Two Windows: Spectral Bounds, Conditioning, and Reproducible Certificates

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Scope. This note certifies two-sided bounds on the spectral infimum of the Weil explicit-formula quadratic form restricted to a compact window, at two explicit windows, together with per-sector conditioning results and one eigenvalue count. Every decisive inequality is produced in Arb ball arithmetic from exact rational inputs and is replayed by a release gate shipped with the package (Section 11).

Not claimed. No arithmetic estimate, no zero-free region, no verified range of zeros, no statement about the Riemann hypothesis, and no result at any window larger than those stated. No priority is claimed over the literature cited in Section 1.3. The certified lower bounds do not extend above their stated windows.

Excluded. An earlier compact time-domain evaluator omitted the truncation error of its outer panelled Gauss–Legendre rule. That route is *excluded* from this release and every quantity depending on it remains uncertified, retained only as a labelled measurement (Section 4.5). Nothing in the results below uses it.

Abstract

Let Q be the Weil explicit-formula quadratic form restricted to test functions supported in a compact window $[-L, L]$, in one fixed normalization, and let $\lambda_Q(L)$ be its spectral infimum. This note reports computer-assisted two-sided bounds at two windows,

$$1.43300048827856 \times 10^{-17} \;\le\; \lambda_Q(4/5) \;\le\; 1.674 \times 10^{-17},$$

$$8.27343868254787 \times 10^{-18} \;\le\; \lambda_Q(\widetilde{L}) \;\le\; 9.475 \times 10^{-18}, \qquad \widetilde{L} = \tfrac{80471895621705018}{10^{17}} < \tfrac{\log 5}{2}.$$

Each lower bound comes from a frequency-cutoff comparison at $T = 400$, reduced to a 440-dimensional Legendre block whose positive definiteness is verified by a preconditioned congruence in Arb ball arithmetic, together with a super-exponentially small complement estimate and a strict Schur-type coupling test; both parity sectors are certified at each window. Each upper bound is the Rayleigh quotient of the *original* form at one explicit even rational test function, evaluated in closed form through the digamma, Lerch Φ and Hurwitz ζ functions — a route with no quadrature, hence no truncation term to bound. Because the form restricts consistently to smaller windows, λ_Q is non-increasing in L; the first lower bound therefore holds throughout $0 < L \le 4/5$, the second throughout $0 < L \le \widetilde{L}$, and the second upper bound for every $L \ge \widetilde{L}$. Pairing the four endpoints gives a corollary on the *true* minima, $0.494 \le \lambda_Q(\widetilde{L})/\lambda_Q(4/5) \le 0.662$. At $L = 4/5$ we also certify, per parity sector, that removing four explicit rational directions from the even sector raises the infimum to at least 0.6175, and that exactly four even eigenvalues lie below 10^{-2} — at most four from the conditioning certificate, at least four from a certified four-dimensional test space for Q

itself. In the odd sector only "at most three" is certified. A proved normalization dictionary identifies this form with the window form of Andrews–Connes–La Gatta under $\lambda = e^L$, verified numerically to about thirty digits. All constants are published as midpoint–radius enclosures with a release gate that replays the chain from the packaged files. This is a computation at two windows: it establishes no arithmetic estimate, nothing about the location of any zeta zero, and nothing about the Riemann hypothesis.

1 Introduction

1.1 The object

Weil's explicit formula pairs a test function against the zeros of ζ, the pole at $s = 1$, the archimedean place and the primes. When the test function is required to be supported in a fixed compact window, the resulting quadratic form becomes a concrete object: a bounded-below Hermitian form on $L^2(-L, L)$ whose positivity *for every* L is equivalent to the Riemann hypothesis [9, 2], and whose positivity at one fixed L is not. Finite-window Weil positivity in this sense is the object studied in the spectral programs of Connes and van Suijlekom [5], Connes and Consani [3], and Connes, Consani and Moscovici [4], and quantitatively in Zhu [10], which supplies the finite-reduction strategy that the present computation follows.

The quantity computed here is the spectral infimum

$$\lambda_Q(L) \;=\; \inf_f \frac{Q_{\zeta,L}(f)}{\|f\|_2^2}$$

of that form at $L = 4/5$, in the normalization fixed in Section 2. Two things make a two-sided statement possible at this window and difficult at larger ones. An *upper* bound needs a single admissible test function and is unconditional. A *lower* bound needs the whole window, hence a reduction of an infinite-dimensional positivity question to a finite matrix plus certified tails; the cost of that reduction grows very fast in L, which is the content of Theorem 1.4 of [10].

1.2 What this note reports

1. A certified lower bound $Q(f) \geq 1.4330 \times 10^{-17} \|f\|_2^2$ at $L = 4/5$, in both parity sectors and hence for arbitrary complex tests (Section 3).

2. A certified upper bound $\lambda_Q(4/5) \leq 1.674 \times 10^{-17}$ from one explicit even rational test function, with the *original* form evaluated in closed form (Section 4). A time-domain candidate is reported but withdrawn pending a quadrature bound (Section 4.5).

3. A certified parity separation: the odd sector satisfies $\lambda_{Q,\mathrm{odd}} \geq 1.3856 \times 10^{-14}$, which is strictly above the upper bound, so the even infimum determines λ_Q (Section 5).

4. Cutoff diagnostics at $T \in \{200, 400, 800\}$ that measure what the comparison discards on fixed test functions, and a certified ceiling on what a larger cutoff can deliver (Section 6).

1.3 Relation to prior work, and what is not claimed

The finite-to-infinite reduction strategy — a scalar gate on the exterior of a frequency cutoff, a Legendre/Bessel representation of the interior, and super-exponentially small tails — is Theorem 1.1 of [10]. Zhu describes supplementary source code and certificates in [10, §17]; these were not obtained or executed in the present work, so his matrix step is neither re-run nor verified here. The reduction was instead rederived in the normalization of the present repository, and every constant below was computed from scratch. The bracket $8.9 \times 10^{-18} \leq \lambda_{\min}(0.8) \leq 2.27 \times 10^{-17}$ stated in the abstract of [10] was used only as an external comparison point and never as an

input, an expected answer, or a tolerance. The parity structure is likewise not new, and at this window it is not merely the decoupling that is already published: [10, Thm. 6.2] certifies the *separation* at support 1.6 — that is, at $L = 4/5$ — and does so by a different route, and establishes more there than this note does, concluding in addition that the ground state is simple and even. What Section 5 contributes is an independent re-derivation of the ordering from this pipeline's own certified data, nothing structural.

The computation continues the implementation developed alongside our finite-window sensitivity study [11], whose repository supplies the zeta-window normalization of Section 2, the geometric-side assembly, the pole conventions, and the closed-form archimedean evaluator used as the validation reference in Section 4(a). The even/odd pole signs used here are the ones checked entrywise against that assembly in our real-zero parity study [12]. Neither the δ^2–L^3 sensitivity estimate of [11] nor the planted-real-zero analysis of [12] is an input to the lower bound proved here: the first concerns a counterfactual perturbation of the zero set, and the second a configuration that is nowhere assumed to exist. The finite-to-infinite reduction follows [10]; what is contributed here is its execution in certified arithmetic, together with the resulting bracket, the numerical parity separation of Section 5, and the cutoff diagnostics of Section 6.

What follows is deliberately narrow, and the following are *not* claimed. No arithmetic estimate is obtained: nothing here improves any bound on primes, on $\Lambda(n)$, on $\psi(x)$, or on a zero-free region. No statement is made about the location of any zero of ζ, and no range of zeros is verified. No new positivity framework is introduced; the framework is that of the cited sources, and the contribution is a certified computation inside it. The computation is at two windows, $L = 4/5$ and $\widetilde{L}$, and gives nothing above $\widetilde{L}$ except the monotone upper bound of Proposition 5 — there is no propagation theorem here, and a certificate at one L does not provide one. Nothing here bears on the Riemann hypothesis. This is a technical note, not peer-reviewed research, and it is not claimed that anyone else uses or has used this work.

1.4 The same window, compared

Because [10] certifies at support 1.6, its results and the present ones describe the same object in the same normalization (its equations (2)–(3) are (1)–(2) below), so they can be put side by side. Doing so is the fairest summary of what this note adds and what it does not.

quantity at $L = 4/5$	[10], certified	this note
whole-window infimum	$[8.9 \times 10^{-18},\ 2.27 \times 10^{-17}]$	$[1.4330 \times 10^{-17},\ 1.674 \times 10^{-17}]$
odd-sector lower bound	8.206×10^{-15}	1.3856×10^{-14}
second even min–max value	$\geq 2.085 \times 10^{-12}$	not certified here
simplicity of the ground state	[10, Thm. 6.2]	not independently established here

Two rows improve on the published result. The width of the whole-window bracket falls from 1.38×10^{-17} to 2.4013×10^{-18}: approximately 82.6% narrower, a reduction by a factor of 5.747, with the present bracket strictly nested inside the published one, both endpoints moving inward. The odd-sector constant is larger by a factor 1.68. Neither is a new method: both are sharper constants obtained inside the framework of [10], from the same kind of certificate. In the remaining two rows [10] is stronger, and the simplicity and evenness of the ground state at this window — which this note does not establish — are his.

As a consistency check — not as an endpoint, and not as a replay of his certificate — [10] also reports a numerical reference estimate $\lambda_{\min}(Q|_{\text{window}}) = 1.656 \times 10^{-17}$ for this window, from his own exact time-domain assembly. That value lies inside the bracket certified here. It is a non-certified reference value in [10] and is used here only in that capacity.

2 Setup, conventions, and the statement

2.1 Normalization

Fix $L > 0$. For $f \in L^2(-L, L)$, extended by zero, write

$$F(t) \;=\; \int_{-L}^{L} f(u)\, e^{itu}\, du, \qquad \text{so} \qquad \int_{\mathbb{R}} |F(t)|^2\, dt \;=\; 2\pi \|f\|_2^2.$$

The window Weil form used throughout is the Hermitian form

$$Q(f) \;=\; 2\,\mathrm{Re}\big(F(i/2)\,\overline{F(-i/2)}\big) \;+\; \frac{1}{2\pi}\int_{\mathbb{R}} |F(t)|^2\, \Psi_L(t)\, dt, \tag{1}$$

$$\Psi_L(t) \;=\; H(t) - P_L(t), \qquad H(t) \;=\; \mathrm{Re}\,\psi\Big(\tfrac{1}{4} + \tfrac{it}{2}\Big) - \log\pi, \qquad P_L(t) \;=\; \sum_{\log n < 2L} \frac{2\Lambda(n)}{\sqrt{n}}\, \cos(t \log n). \tag{2}$$

Here ψ is the digamma function and Λ is the von Mangoldt function. Writing $C = \int f \cosh(u/2)\, du$ and $S = \int f \sinh(u/2)\, du$, so that $F(\mp i/2) = C \pm S$, the pole term of (1) is

$$2\,\mathrm{Re}\big(F(i/2)\,\overline{F(-i/2)}\big) \;=\; 2\big(|C|^2 - |S|^2\big), \tag{3}$$

which is the form that must be used for complex f: it is additive over the real and imaginary parts, whereas $2F(i/2)^2$ is not — replacing f by if reverses its sign. The computation never needs (3) in that generality. It is carried out on *real* test functions of definite parity, where C survives on even f and S on odd f, and (3) specializes to

$$2s\, F(i/2)^2, \qquad s = +1 \text{ (even)}, \qquad s = -1 \text{ (odd)},$$

the two signs recording that the pole helps the even sector and hurts the odd one. Arbitrary complex f is then reached through the decomposition of Section 3.9, not through (3) directly.

The two sector signs are not asserted here: in the parity study [12, §3(c)] the $\delta = 1/2$ real-pair perturbation is compared entrywise with this repository's own pole term $2\,\mathrm{pole_sign}(\varepsilon) P_j P_k$ and agrees in all nine tested cells to 6.22×10^{-61}, with $\mathrm{pole_sign} = -1$ on the odd sector, while the opposite sign convention is wrong by $O(1)$ to $O(10)$; and [12, Remark 6] identifies that $\delta = 1/2$ case as the parity splitting of [10, Lemma 6.1]. The prime sum in (2) is *finite and exact*: the autocorrelation of a function supported in $[-L, L]$ is supported in $[-2L, 2L]$, so only n with $\log n < 2L$ contribute and there is no truncation. At $L = 4/5$ the active prime powers are $\{2, 3, 4\}$.

This is the normalization of `src/zeta_window.py` in the repository accompanying this note, documented in [11], where the same object is written $T_R = A - 2M$ with $A = \mathrm{Pole} + \mathrm{Arch} - (\log\pi) I$ and $2M = \mathrm{Prime}$. It is an imported convention, not derived here, and the grouping of $-\log\pi$ is part of it. It was cross-checked against the repository's own independent closed form for the constant test function (Section 4, item (a)).

2.2 The admissible class

Since H is even, continuous, and increasing on $t > 0$ (Lemma 8), and $|P_L| \le A_L := \sum_{\log n < 2L} 2\Lambda(n)/\sqrt{n} < \infty$, the function Ψ_L is bounded below on $\mathbb{R}$ and bounded on compacts. Hence for *every* $f \in L^2(-L, L)$ the negative part of the integral in (1) is finite and $Q(f)$ is well defined in $(-\infty, +\infty]$. We therefore take the admissible class to be all of $L^2(-L, L)$, with the convention that a lower bound $Q(f) \ge \mu \|f\|_2^2$ is satisfied trivially where $Q(f) = +\infty$. This is a choice of convention made in this note; it strengthens rather than weakens the statement, and it removes any need for a density or form-domain argument.

2.3 Three spectral quantities, kept apart

Confusing the following three numbers is the easiest way to overstate a result of this kind, so they are named separately and used separately throughout.

Definition 1. With $T > 0$ a frequency cutoff, β a constant with $\Psi_L(t) \geq \beta$ for $|t| \geq T$, and V_N a finite-dimensional subspace of $L^2(-L, L)$:

$$\begin{aligned} \lambda_Q &= \inf\{\, Q(f)/\|f\|_2^2 \,\} && \text{over the admissible class,} \\ \lambda_R(T) &= \inf\{\, R_{T,\beta}(f)/\|f\|_2^2 \,\} && \text{over the same class, } R_{T,\beta} \text{ as in (4),} \\ \lambda_A(T, N, \varepsilon) &= \lambda_{\min}(\text{compression of } R_{T,\beta} \text{ to } V_N) && \text{in the parity sector } \varepsilon. \end{aligned}$$

Because $Q \geq R_{T,\beta}$ pointwise, and because an infimum over a subspace is at least the infimum over the whole space,

$$\lambda_R(T) \;\leq\; \lambda_Q, \qquad \lambda_R(T) \;\leq\; \lambda_A(T, N, \varepsilon),$$

and there is **no** inequality in either direction between λ_A and λ_Q. In particular a numerical least eigenvalue of a finite block is not an upper bound for λ_Q, and only the evaluation of Q itself on an explicit test function is.

2.4 The theorem

Theorem 2. *Let Q be the form (1)–(2) on $L^2(-L, L)$ and $\lambda_Q(L) = \inf\{Q(f)/\|f\|_2^2\}$. Put*

$$\widetilde{L} \;=\; \frac{80471895621705018}{10^{17}}, \qquad \textit{so} \qquad \tfrac{4}{5} \;<\; \widetilde{L} \;<\; \frac{\log 5}{2}.$$

(i) (Lower bound at $L = 4/5$.) *For every $f \in L^2(-4/5, 4/5)$,*

$$Q(f) \;\geq\; 1.43300048827856 \times 10^{-17}\, \|f\|_2^2,$$

in both parity sectors separately and hence for arbitrary complex f.

(ii) (Upper bound at $L = 4/5$.) *There is an explicit even test function f_{160}, a combination of the 160 even cosine modes $\cos(k\pi u/L)$, $k = 0, \ldots, 159$, with rational coefficients of denominator 10^{180}, such that*

$$\lambda_Q(4/5) \;\leq\; \frac{Q(f_{160})}{\|f_{160}\|_2^2} \;\leq\; U_0 \;=\; 1.673125534070681868077268377 36\times 10^{-17} \;<\; 1.674\times 10^{-17}.$$

(iii) (Lower bound at $\widetilde{L}$.) *For every $f \in L^2(-\widetilde{L}, \widetilde{L})$,*

$$Q(f) \;\geq\; 8.27343868254787 \times 10^{-18}\, \|f\|_2^2,$$

again in both parity sectors separately.

(iv) (Upper bound at $\widetilde{L}$.) *There is an explicit even test function $\widetilde{f}_{160}$ of the same shape such that*

$$\lambda_Q(\widetilde{L}) \;\leq\; U_- \;=\; 9.474769510830490328497 09 \times 10^{-18} \;<\; 9.475 \times 10^{-18}.$$

(v) (Parity separation at $L = 4/5$.) *For every odd $f \in L^2(-4/5, 4/5)$, $Q(f) \geq 1.38569999999861\times 10^{-14}\|f\|_2^2$. Since f_{160} is even and $1.3856 \times 10^{-14} > U_0$, the even infimum equals $\lambda_Q(4/5)$.*

Parts (i), (iii) and (v) are computer-assisted: they rest on interval arithmetic in the Arb midpoint–radius model [7], including all analytic error terms, on exact rational input data, and on a release gate that recomputes every decisive implication from the packaged files. Parts (ii) and (iv) each need one test function and are unconditional given the normalization.

Remark 3. The shortened decimals above are directed: an upper bound is quoted rounded *up* and a lower bound rounded *down*. In particular 1.674×10^{-17} is a valid short form for U_0 while 1.6731×10^{-17} is *not*, being below U_0; and 1.4330×10^{-17} is a valid short form for the constant in (i). Section 11 records the check that enforces this on every printed bound.

Remark 4. Theorem 2(v) identifies which sector carries the infimum. It does *not* assert that the infimum is attained, that a minimizer is unique, or that the least eigenvalue is simple.

Proposition 5 (Monotonicity and coverage)**.** *If $0 < L' \le L$ then every f supported in $[-L', L']$ is admissible for the window $[-L, L]$ and Q takes the same value on it, so $\lambda_Q(L) \le \lambda_Q(L')$: the infimum is non-increasing in L. Consequently Theorem 2 gives*

$$\lambda_Q(L) \;\ge\; 1.43300048827856\times10^{-17} \quad (0 < L \le \tfrac{4}{5}), \qquad \lambda_Q(L) \;\ge\; 8.27343868254787\times10^{-18} \quad (0 < L \le \widetilde{L}),$$

$$\lambda_Q(L) \;\le\; 9.475 \times 10^{-18} \quad (L \ge \widetilde{L}), \;\; \textit{in particular at } L = \tfrac{\log 5}{2}.$$

No positive lower bound at $L = (\log 5)/2$ follows from these certificates: monotonicity runs the other way.

Corollary 6 (Change between the two minima)**.** *With μ_0, U_0 and μ_-, U_- the certified lower and upper endpoints at $4/5$ and $\widetilde{L}$,*

$$\frac{\mu_-}{U_0} \;\le\; \frac{\lambda_Q(\widetilde{L})}{\lambda_Q(4/5)} \;\le\; \frac{U_-}{\mu_0}, \qquad \textit{hence} \qquad 0.494 \;\le\; \frac{\lambda_Q(\widetilde{L})}{\lambda_Q(4/5)} \;\le\; 0.662,$$

so the spectral infimum falls by between 33.8% and 50.6% between the two windows.

Proof. Both brackets are two-sided, and $\lambda_Q > 0$ at both windows by Theorem 2(i),(iii), so the quotient is bounded by pairing the lower endpoint at each window with the upper endpoint at the other. Taking the four packaged exact decimals $\mu_0 = 1.43300048827856 \times 10^{-17}$, $U_0 = 1.674 \times 10^{-17}$, $\mu_- = 8.27343868254787 \times 10^{-18}$, $U_- = 9.475 \times 10^{-18}$ gives $\mu_-/U_0 = 0.4942317\ldots$ and $U_-/\mu_0 = 0.6612000\ldots$, and the displayed bounds are those rounded outward. The divisions are exact rational arithmetic. □

Remark 7. Corollary 6 is a consequence of the two brackets and nothing else: no prolate quantity, no parameter transfer, and no numerical eigenvalue from the literature enters it. It is *not* an independent analytic scaling theorem, and it does not establish any asymptotic law. See Section 10 for what it does and does not say about Hypothesis 3.4 of [1].

3 The certified lower bound

The lower bound is assembled in five steps. Steps 1–5 are proved here in the normalization of Section 2; the strategy is that of [10, Thm. 1.1].

3.1 Step 1: the exterior gate

Lemma 8. *$t \mapsto \operatorname{Re}\psi(\frac{1}{4} + \frac{it}{2})$ is strictly increasing on $t > 0$.*

Proof. From the digamma series,

$$\operatorname{Re}\psi\Big(\tfrac{1}{4} + \tfrac{it}{2}\Big) = -\gamma + \sum_{n\ge0}\left[\frac{1}{n+1} - \frac{n+1/4}{(n+1/4)^2 + t^2/4}\right],$$

and each summand is strictly increasing in $t > 0$. Termwise differentiation is justified by local uniform convergence. □

Consequently, with $A_L = \sum_{\log n<2L} 2\Lambda(n)/\sqrt{n}$ and $\beta := H(T) - A_L$, we get $\Psi_L(t) \geq \beta$ for all $|t| \geq T$, because $P_L \leq A_L$ pointwise. At $L = 4/5$,

$$A_L = 2.941973525223620455503909 \pm 3.91 \times 10^{-25},$$

and the gate values at the three cutoffs used in this note are

T	$\beta = H(T) - A_L$	status
200	$0.518465733243867199873536 \pm 2.62 \times 10^{-26}$	certified
400	$1.211613695058064549471502 \pm 1.89 \times 10^{-26}$	certified
800	$1.904761070930776888621206 \pm 1.68 \times 10^{-25}$	certified

with the increments $\beta_{400} - \beta_{200} = 0.693147961814217829484148 4 \pm 7.24 \times 10^{-27}$, numerically close to $\log 2$ as the asymptotics of H suggest. All three are *positive*, so the set $\{|t| \geq T : \Psi_L(t) < \beta\}$ is empty and no occupation or measure estimate on a bad exterior set is needed anywhere below.

3.2 Step 2: the comparison form

Because $\int_{|t|\geq T}(\Psi_L - \beta)|F|^2 \geq 0$, equation (1) gives $Q(f) \geq R_{T,\beta}(f)$ with

$$R_{T,\beta}(f) \;=\; \beta\|f\|_2^2 \;+\; 2sF(i/2)^2 \;+\; \frac{1}{2\pi}\int_{-T}^{T} (\Psi_L(t) - \beta)|F(t)|^2\, dt. \tag{4}$$

The discarded quantity is exactly

$$e_T(f) \;:=\; \frac{Q(f) - R_{T,\beta}(f)}{\|f\|_2^2} \;=\; \frac{1}{2\pi\|f\|_2^2}\int_{|t|>T} (\Psi_L(t) - \beta)|F(t)|^2\, dt \;\geq\; 0, \tag{5}$$

which is measured in Section 6.

Proposition 9 (monotonicity in β)**.** *By Plancherel, $R_{T,\beta}(f) = 2sF(i/2)^2 + \frac{1}{2\pi}\int_{|t|\leq T}\Psi_L|F|^2 + \frac{\beta}{2\pi}\int_{|t|>T}|F|^2$, so for $\beta_2 \geq \beta_1$,*

$$R_{T,\beta_2}(f) - R_{T,\beta_1}(f) = \frac{\beta_2 - \beta_1}{2\pi}\int_{|t|>T}|F|^2 \;\geq\; 0.$$

On the finite compression this reads $A(\beta_2) - A(\beta_1) = (\beta_2 - \beta_1)(I - K_T)$, where K_T is the compression of the Fourier energy-concentration operator and $0 \leq K_T \leq I$.

Proposition 9 is recorded because it rules out a tempting move: *lowering* β cannot raise the exact comparison form, its infimum, or the least eigenvalue of its compression to a fixed subspace. Inspecting only the factor $\Psi_L - \beta$ inside the integral misses the $\beta\|f\|_2^2$ term. The identity was checked numerically with maximum residual 5.23×10^{-143} and diagonal entries of K_T in [0.9420888637695, 0.9980132600038] at $T = 200$, $N = 16$.

3.3 Step 3: the Legendre basis

Let $\varphi_n(u) = \sqrt{(2n+1)/(2L)}\, P_n(u/L)$ be the orthonormal Legendre basis of $L^2(-L, L)$. Then

$$F_n(t) \;=\; i^n\sqrt{2L(2n+1)}\; j_n(Lt), \qquad \sum_{n\geq 0}(2n+1)j_n(x)^2 = 1,$$

with j_n the spherical Bessel function. The form is assembled in the real coordinates $y_n = \sigma_n x_n$ with $\sigma_n = (-1)^{\lfloor n/2\rfloor}$, which makes $v_n(t) = \sqrt{2L(2n+1)}\, j_n(Lt)$ the real working transform. This basis is used, rather than the cosine basis of the repository's original assembly, because $j_n(x)$ is super-exponentially small for $n > ex/2$, which is what makes Step 4 effective; a cosine basis gives only $O(n^{-2})$ tail decay and cannot reach the required magnitude at an acceptable dimension.

3.4 Step 4: the complement

For $f \in V_N^{\perp}$ with $\|f\|_2 = 1$, Cauchy–Schwarz in n gives $|F(t)|^2 \le 2L\, S_N(tL)$ with $S_N(x) = \sum_{n \ge N} (2n+1) j_n(x)^2$. Since S_N is increasing on $0 \le x < N$,

$$g_N \;:=\; \sup_{f \in V_N^{\perp},\, \|f\|=1} \frac{1}{2\pi} \int_{-T}^{T} |F|^2 \;\le\; \frac{2LT}{\pi}\, S_N(TL).$$

This is a lossy but rigorous surrogate for the $(N+1)$-st Slepian eigenvalue [8], and it is super-exponentially small past the Shannon number $2TL/\pi$. At $T = 400$, $N = 440$:

$$S_N(320) = 9.260051461963608 \times 10^{-63}, \qquad g_N = 1.886442193224756 \times 10^{-60}.$$

3.5 Step 5: the Schur-type block bound, with strict tests

Split $f = f_N + f_\perp$. With a a certified lower bound for the least eigenvalue of the compression of $R_{T,\beta}$ to V_N, d a certified lower bound for $R_{T,\beta}$ on $V_N^{\perp}$, and κ a certified bound on the coupling, a 2×2 comparison yields positivity as soon as $ad > \kappa^2$. The threshold κ^2/d is a *break-even point*, not the resulting bound. To avoid the cancellation-prone explicit eigenvalue formula, the final implication is tested as three *strict* interval signs for the proposed constant μ:

$$a > \mu, \qquad d > \mu, \qquad (a-\mu)(d-\mu) > \kappa^2. \tag{6}$$

Equality anywhere in (6) would give only a nonnegative comparison, so strictness is required and is checked as a sign of an interval, never of a midpoint. The coupling constant is $\kappa = 2|p_N||p_\perp| + M\sqrt{g_N}$ with $M = \sup_{|t| \le T} |\Psi_L - \beta|$ and p the pole vector $p_n = \sqrt{2L(2n+1)}\, i_n(L/2)$ taken in the same y-coordinates; $|p|^2 = \int_{-L}^{L} e^{-u}\, du = 2 \sinh L$ exactly. M is obtained from the monotonicity of H alone, with no covering argument.

3.6 Verified positive definiteness

A naive interval LDL^{T} factorization is unusable at this dimension: entry radii of order 10^{-38} amplify by roughly 6.5 per pivot and reach $O(1)$ by step 47. Instead, positive definiteness of $B = A - aI$ is established by a preconditioned congruence in the sense of Rump [6]. Let R be the upper triangular Cholesky factor of the *midpoint* of B, treated as exact input, and set $C = R^{-1}$ and

$$G \;=\; C^* B\, C \;=\; R^{-\mathsf{T}} (A - aI)\, R^{-1},$$

computed by two explicit forward substitutions in ball arithmetic. The verified inequality is

$$\|I - G\|_\infty \;<\; 1.$$

This proves $B \succ 0$: the bound places every eigenvalue of the Hermitian G in $(0, 2)$, so $G \succ 0$; if C were singular then $G = C^*BC$ would be singular, so C is nonsingular; and then for $x \neq 0$, writing $x = Cy$ with $y \neq 0$, $x^*Bx = y^*Gy > 0$. It is a congruence, not a one-sided inverse residual — the latter would establish nonsingularity only.

3.7 Quadrature

The interior integral in (4) is evaluated by panelled Gauss–Legendre quadrature with certified nodes and weights (FLINT's `legendre_p_root`). The symbol, prime comb and the full basis-transform vector are evaluated once per node and reused across all matrix entries, so no entry is integrated adaptively. The quadrature remainder is bounded by a Bernstein-ellipse estimate on genuinely holomorphic extensions: the half-sum

$$H_{\mathrm{hol}}(z) \;=\; \tfrac{1}{2}\left[\psi\!\left(\tfrac{1}{4} + \tfrac{iz}{2}\right) + \psi\!\left(\tfrac{1}{4} - \tfrac{iz}{2}\right)\right] - \log \pi$$

agrees with H on the real axis and has its nearest poles at $z = \pm i/2$ (note that $\operatorname{Re}\psi(\frac{1}{4} + \frac{iz}{2})$ is *not* a holomorphic extension), and the basis product is extended as $v_n(z)v_m(-z)$ rather than by conjugation at a complex argument. On the ellipses, $|v_n(z)|$ is bounded uniformly in n by

$$\Big|\int_{-L}^{L} \varphi(u)e^{izu}\,du\Big| \;\le\; \Big(\int_{-L}^{L} e^{-2u\operatorname{Im} z}du\Big)^{1/2} \;\le\; \sqrt{2L}\,e^{L|\operatorname{Im} z|} \tag{7}$$

for any L^2-normalized φ supported in $[-L, L]$. Inequality (7) is immediate from Cauchy–Schwarz and is *not* claimed as new; it is recorded because it replaces a direct interval evaluation of j_n on wide complex balls, which is numerically useless here. The remainder is folded into the matrix entries as a widening exactly once.

3.8 The $T = 400$ certificate

parameter	value
window	$L = 4/5$
cutoff	$T = 400$
dimension	$N = 440$ (Legendre degrees 0..439; 220 modes per parity)
quadrature	800 panels of half-length 1/4, Gauss–Legendre order 64
Bernstein data	$b_e = 0.30$ (inside the strip $\lvert\operatorname{Im} z\rvert < 1/2$), $\rho = 2.762049935$
remainder factor	$3.551349410 \times 10^{-57}$
precision	900 bits (Bessel), 384 bits (accumulation), 1400 bits (verification)

$N = 440$ was chosen from the actual inequalities, not from the heuristic $eLT/2 = 434.9$. The certified constants are

$$\beta = 1.21161369505806454947150 2, \qquad M = 9.52577063950735058720 8369,$$
$$\kappa = 1.30834377128538825902 9018 \times 10^{-29}, \qquad \kappa^2/d = 1.412796364751588 \times 10^{-58},$$

with pole tail $|p_\perp| = 7.993372152149564 \times 10^{-1282}$.

sector	a (certified shift)	$\|I - G\|_\infty$
even	$1.43300048828 \times 10^{-17}$	$1.1201203 \times 10^{-12}$
odd	1.3857×10^{-14}	$5.6557966 \times 10^{-18}$

sector	d	μ (published)	margin in (6)
even	1.211613695058065	$1.43300048827856 \times 10^{-17}$	$1.74472372088 \times 10^{-29}$
odd	1.211613695058065	$1.38569999999861 \times 10^{-14}$	$1.68414303613 \times 10^{-26}$

The whole-window constant is the smaller of the two sector constants, so $\mu = 1.43300048827856 \times 10^{-17}$ and *the even sector binds*. Assembly took 3532 s, the full run 3581 s, with peak resident memory 393.4 MiB; the two Rump factorizations at $N = 440$ took under 50 s together.

3.9 Parity and complex tests

The form decouples by parity: the interior matrix has $C_{nm} = 0$ when $n + m$ is odd, because $v_n v_m$ is then odd on a symmetric interval, and the pole term $2(C^2 - S^2)$ separates into its even and odd parts. For complex $f = u + iv$ the Hermitian pole term (3) gives $2(|C|^2 - |S|^2) = 2(C_u^2 - S_u^2) + 2(C_v^2 - S_v^2)$, since $|C|^2 = C_u^2 + C_v^2$ and $|S|^2 = S_u^2 + S_v^2$. The multiplier term splits as well, but *not* pointwise: with $F = F_u + iF_v$,

$$|F(t)|^2 \;=\; |F_u(t)|^2 + |F_v(t)|^2 \;-\; 2\operatorname{Im}\Big(\overline{F_u(t)}\,F_v(t)\Big)\,,$$

and the cross term does not vanish identically. It is, however, *odd* in t: for real u one has $F_u(-t) = \overline{F_u(t)}$, so $\overline{F_u(-t)}F_v(-t) = \overline{F_u(t)F_v(t)}$ and its imaginary part changes sign. Since Ψ_L is even, the cross term integrates to zero against it, and $\frac{1}{2\pi}\int_{\mathbb{R}} |F|^2\Psi_L = \frac{1}{2\pi}\int_{\mathbb{R}}(|F_u|^2 + |F_v|^2)\Psi_L$. Hence $Q(f) = Q(u) + Q(v)$; each real part then splits by parity. Hence the two sector certificates cover the full class, which is what makes Theorem 2(i) a statement about arbitrary complex f.

Remark 10. A positive-definiteness check cannot serve as an oracle for the pole sign. The direct convention checks of Section 4 are retained for that purpose, because an earlier version of this computation carried a pole-phase error that produced a plausibly negative pivot (Appendix A, item 1).

4 The original-form upper bound

4.1 The certified bound, from the closed-form route

A variational upper bound needs one admissible test function and an evaluation of Q on it that is certified end to end. The evaluation used for Theorem 2(ii) is the *closed-form* one: in the even cosine basis $f_k(u) = \cos(k\pi u/L)$ the prime sum is finite and exact, and the archimedean term is summed analytically through the digamma, Lerch Φ and Hurwitz ζ functions, each evaluated as an Arb ball. **This route performs no quadrature of any kind**, so it carries no quadrature truncation error to bound — which is exactly why it, and not the sharper route of Section 4.2, supplies the certified endpoint.

The procedure has two stages, which are kept apart. An exploratory floating-point eigensolver *locates* a candidate vector in this basis; it contributes no certified claim, and a poor candidate costs a weaker bound, never a wrong one. The candidate is then frozen to exact rationals and its Rayleigh quotient enclosed in Arb inside an owned precision scope. Because the second stage is exact, the first needs only enough precision to find a good trial function, which is why the mode count can be pushed without an expensive eigensolve.

Raising the mode count lowers the bound, and the sequence is the useful record:

N	certified $Q(f_N)/\|f_N\|_2^2 \le$	
28	$2.1129926117214908 \times 10^{-17}$	(the phase-1 witness)
48	$1.9204310132 \times 10^{-17}$	
64	$1.8541624482 \times 10^{-17}$	
96	$1.7476544722 \times 10^{-17}$	
128	$1.6972723502 \times 10^{-17}$	
160	$1.6731255341 \times 10^{-17}$	used in Theorem 2(ii)

The witness f_{160} is a frozen exact rational vector over the 160 even modes $k = 0, \dots, 159$ with denominator 10^{180}. Re-certified from those exact rationals alone — no eigensolver, no floating point — at 2920 bits,

$$\frac{Q(f_{160})}{\|f_{160}\|_2^2} \;=\; 1.67312553407068186807726837736 \times 10^{-17} \pm 4.41 \times 10^{-47},$$

whose outward upper endpoint — the midpoint plus the radius, not the printed midpoint itself — is below the short certified form

$$U \;\le\; 1.674 \times 10^{-17} \;<\; 1.674 \times 10^{-17},$$

giving Theorem 2(ii). The radius is a consequence of the route: with no quadrature and no truncation, only ball rounding contributes. For comparison, [10] reports a numerical reference estimate 1.656×10^{-17} for this infimum (Section 1.4), so the certified endpoint above sits about 1% over it.

Remark 11 (on quoting an endpoint). A serialized enclosure must be consumed through the correct endpoint — lower for a lower bound, upper for an upper bound — and never through its midpoint or its printed decimal. Two earlier drafts of this note violated that: one quoted an enclosure's midpoint as the bracket's upper endpoint, and one quoted an endpoint obtained by re-parsing a stored ball at the ambient 53 bits. Both are corrected (Appendix A, item 6), and the safe decimals 1.674×10^{-17} here and 1.4330×10^{-17} in part (i) are the forms in which the result should be quoted. The 30-digit decimal above is the *midpoint* of the stored enclosure (`phase5/certs/upper_highN.json`, field `enclosure`); its outward upper endpoint exceeds that midpoint by at most the stored radius 4.41×10^{-47}, and the artifact's own `safe_decimal` field records 1.674×10^{-17} as the short form that dominates it. Quoting the midpoint as though it were the endpoint is exactly the error of item 6, and is avoided here.

4.2 A sharper candidate: the time-domain route

A variational upper bound requires Q itself, not $R_{T,\beta}$. Evaluating (1) directly would mean certifying an oscillatory integral over all of $\mathbb{R}$. Instead Q is evaluated on the geometric side via the autocorrelation

$$g(x) \;=\; \int f(s+x)f(s)\,ds,$$

supported in $[-2L, 2L]$, together with the compact archimedean identity ([10, Lemma 2.5], transcribed into the present normalization)

$$\mathrm{Arch}(g) = -\big[\gamma_E + \log\pi + \log(1-e^{-4L})\big]g(0) + \int_0^{2L} \frac{2\big[e^{-2x}g(0) - e^{-x/2}g(x)\big]}{1-e^{-2x}}\,dx, \tag{8}$$

$$Q(f) = 2sF(i/2)^2 + \mathrm{Arch}(g) - \sum_{\log n < 2L} \frac{2\Lambda(n)}{\sqrt{n}}\,g(\log n). \tag{9}$$

This is the route by which [10, §6.2] evaluates its own variational upper bounds at this window, there with shift-correlation integrands of degree at most 239 integrated exactly by a 160-node Gauss rule. Two features make it cheap and rigorous. They are set out below because the present error accounting depends on them, not as new observations.

Exact inner quadrature. If f is a polynomial of degree d on $[-L, L]$ then g restricted to $[0, 2L]$ is a polynomial of degree at most $2d+1$ — not $2d$: for $f \equiv 1$ one has $g(x) = 2L - x$. What governs the *inner* rule is the degree of the integrand $f(s+x)f(s)$ in the integration variable s, which is at most $2d$, so an m-node Gauss–Legendre rule is exact once $2m-1 \geq 2d$, i.e. $m \geq d+1$. The implementation uses $m = d+2$; for $d = 438$ that is 440 nodes. *Exactness* here means no polynomial quadrature truncation error in exact arithmetic — node, weight, coefficient and rounding enclosures are still carried as ball radii, and the outer integral in (8) keeps its own certified Bernstein remainder.

The removable endpoint, by an identity. The integrand of (8) is removable at $x = 0$ and its limit involves $c = -g'(0^+)$. For even f,

$$g'(x) \;=\; -f(L)f(L-x) + \int_{-L}^{L-x} f'(s+x)f(s)\,ds,$$

and at $x = 0$ the integral is $\frac{1}{2}[f^2]_{-L}^{L} = 0$, so

$$c \;=\; -g'(0^+) \;=\; f(L)^2 \qquad \text{exactly,} \tag{10}$$

and the integrand limit is $c - \frac{3}{2}g(0)$. This closed form is an implementation convenience obtained by elementary differentiation and integration; [10, Lemma 2.5] leaves c as the one-sided limit

and remarks only that it is nonzero in general. Its absence from the displayed formulas there is not evidence that it is new, and no novelty is claimed for it. The outer integrand is analytic in $|\operatorname{Im} x| < \pi$; panels are geometric near the origin so that no Bernstein ellipse encloses it, and the residual head $[0, 10^{-40}]$ is bounded by an explicit constant.

4.3 Validation of the representation

(a) On the constant $f = \varphi_0$, the compact route gives $Q(\varphi_0) = 0.080274054959269611214 \pm 3.09\times10^{-22}$, matching the repository's independent closed form $Q[0,0] = 0.080274054959269611214$ to all 21 printed digits (difference -3.08×10^{-22}). That closed form is the evaluator of [11], which sums the archimedean series analytically through the digamma, Lerch Φ and Hurwitz ζ functions; it is an independent route to Q, not an independent implementation of the present one.

(b) A degree-0 check cannot see a pole-phase error, since $\sigma_0 = +1$. Checks at φ_2 and at the signed mixed function $\varphi_0 - \varphi_2$ were therefore added, with the pole moment computed *directly* from the physical-space function rather than from the pole-vector helper, and with the finite prime shifts re-enumerated independently of the symbol helper:

f	$Q(f)$	pole moment (direct)
φ_0	0.080274054959269611214	1.29891290374844725
φ_2	0.389336213205013655883	0.030516224225381024
$\varphi_0 - \varphi_2$	0.117009437226589934780	1.26839667952306623

All agree with the helper values, the first to 10^{-25}.

(c) *Disclosure.* This is a second *representation* of the same form, not an independent implementation: it shares the Arb library, the symbol helper for the comb data, and the pole-vector helper. It is not described as an independent check.

4.4 The frozen test functions of the time-domain route

Two trial functions are used, each a frozen exact rational vector in the Legendre y-coordinates with denominator 10^{40}:

- f_{200}: 110 even modes, degrees 0..218, $\|f\|_2^2 = 2.27524444365096536097055 0$;
- f_{400}: 220 even modes, degrees 0..438.

Both were *located* by an exploratory floating-point eigensolver, then rounded to exact rationals and certified; the eigensolver contributes no certified claim, and in particular supplies no eigenvalue.

For f_{200}, equation (9) gives, at 94 outer panels and head cut 10^{-40},

$$\text{pole} = 3.2303213242823875400, \qquad \text{Arch} = -3.1402989129419417722,$$

$$\text{comb} = 0.090022411340445684793,$$

whose signed sum is $8.30143811294573721 \times 10^{-17}$: a cancellation of about sixteen digits, which is why the evaluation is done at high precision in ball arithmetic and why the compact representation — with its exact inner quadrature and the exact identity (10) — matters. Dividing by the norm,

$$q(f_{200}) = 3.64859175290407702 \times 10^{-17} \pm 4.15 \times 10^{-35}.$$

This is above the then-current upper bound and so does not improve it; the expected outcome, since f_{200} minimizes a comparison form, not Q.

For f_{400},

$$q(f_{400}) = 1.90138291114546440 \times 10^{-17} \pm 6.03 \times 10^{-35},$$

which is *above* the certified endpoint U_0 of Section 4.1 and so would not improve the bracket even were the enclosure complete. It is not complete; see Section 4.5. The radii printed above reflect ball rounding and the explicit head bound only.

4.5 Excluded route: the time-domain evaluator

The outer integral in (8) is evaluated by panelled Gauss–Legendre quadrature — geometric panels from $\eta = 10^{-40}$ to 10^{-3}, then 32 uniform panels of order 32 to $2L$ — and the implementation adds the explicit head bound on $[0, \eta]$ but **no bound on the quadrature truncation error of those panels**. Ball arithmetic carries the node, weight and rounding enclosures; it does not capture the difference between a Gauss–Legendre sum and the integral it approximates, which is a property of the rule rather than of the arithmetic. The radii printed in Section 4.2 therefore do not include that term.

Accordingly, **every value produced by the time-domain route is withdrawn as a certificate** pending such a bound: $q(f_{200})$, $q(f_{400})$, the endpoint once quoted from the latter, and the discarded energies $e_T = q - r_T$ of Section 6 together with their ratios. They are retained below as *measurements*, labelled as such.

What is *not* affected, and why:

- the lower bound of Section 3, whose interior quadrature carries a certified Bernstein-ellipse remainder folded into the matrix entries exactly once, re-derived by its verifier;
- the odd-sector bound of Theorem 2(v), from the same matrix route;
- the certified upper bound of Section 4.1, which uses no quadrature at all;
- the scalar values $R_T(f)$ of Section 6, which are computed by a separate routine that *does* add a certified remainder and publishes it; in particular the ceiling $\lambda_R(800) \le r_{800}(f_{400})$ does not involve Q at any point.

A rigorous remainder for (8) requires a certified bound on the analytic continuation of the outer integrand on a Bernstein ellipse around each panel, with the denominator's zeros at $x \in i\pi\mathbb{Z}$ kept outside. The degree statement corrected above is a prerequisite for that bound and not a substitute for it: knowing $\deg g \le 2d + 1$ says nothing by itself about the size of g off the real axis. The obvious route — bound $|g|$ on the ellipse by $\max_{[0,2L]} |g| \cdot R^{\deg g}$ — is far too lossy at $\deg g = 877$, forcing panels of width $\sim 10^{-3}$. A coefficient-sensitive bound should work instead, because the frozen minimizers concentrate on *low* Legendre orders, so estimating $|f(w)|$ off the real axis directly from the frozen coefficients is tight exactly where $R^{\deg g}$ is hopeless. That derivation is left open; it is recorded with the diagnostic in `phase5/src/outer_convergence.py`, which measures the refinement difference at $\pm 2.71 \times 10^{-35}$ — small, and not a bound.

Since the certified endpoint of Section 4.1 now sits *below* the withdrawn time-domain value, closing this gap would no longer improve the bracket; it would restore a second, independent route to it.

5 The parity separation

The $T = 400$ certificate of Section 3 delivers $a = 1.3857 \times 10^{-14}$ in the odd sector against $a = 1.43300048828 \times 10^{-17}$ in the even sector. Re-evaluating the three strict signs (6) from

the exact certified data — not from rounded report values — at the published constant $\mu_{\text{odd}} = 1.38569999999861 \times 10^{-14}$ gives

$$a > \mu_{\text{odd}}, \qquad d > \mu_{\text{odd}}, \qquad (a - \mu_{\text{odd}})(d - \mu_{\text{odd}}) > \kappa^2,$$

with margin $1.68414303613 \times 10^{-26}$. So the whole odd sector satisfies $Q(f) \geq 1.38569999999861 \times 10^{-14}\|f\|_2^2$. Combining with the certified endpoint U_0 of Section 4.1, and using that the witness f_{160} is even (the short form 1.3856×10^{-14} below rounds the published constant *down*, as a lower bound must),

$$\mu_{\text{odd}} - U_0 \;=\; 1.384026874464 \times 10^{-14} \;>\; 0,$$

so

$$\inf_{\text{even}} \frac{Q(f)}{\|f\|_2^2} \;\leq\; U_0 \;<\; \mu_{\text{odd}} \;\leq\; \inf_{\text{odd}} \frac{Q(f)}{\|f\|_2^2},$$

which with the orthogonal parity decomposition identifies the even infimum as $\lambda_Q(4/5)$. The global lower bound is $1.43300048827856 \times 10^{-17}$, governed by the even sector.

The witness used here is the closed-form one. An earlier draft ran this argument on the time-domain value for f_{400}; that value is withdrawn by Section 4.5, and the separation is re-established above on a witness whose evaluation uses no quadrature. The conclusion is unchanged.

As noted in Section 1.3, none of this *structure* is new: at this very window [10, Thm. 6.2] certifies the same ordering, with $\lambda_1^{\text{odd}} \geq 8.206 \times 10^{-15}$. The odd constant recorded above exceeds that value by a factor 1.68, which is a sharpening of one constant and nothing more; the separation itself, and the simplicity and evenness of the ground state at this window, were established by [10, Thm. 6.2] and are not reproduced here. Nothing about simplicity, uniqueness or attainment follows from the numbers above.

6 Cutoff diagnostics

6.1 The cutoff comparison, proved

Proposition 12. *Let $\Psi_L \geq \beta_i$ on $|t| > T_i$ for $i = 1, 2$, with $T_2 \geq T_1$ and $\beta_2 \geq \beta_1$. Then*

$$R_{T_2,\beta_2}(f) - R_{T_1,\beta_1}(f) = \frac{1}{2\pi} \int_{T_1 < |t| \leq T_2} (\Psi_L - \beta_1)|F|^2 + \frac{\beta_2 - \beta_1}{2\pi} \int_{|t| > T_2} |F|^2 \;\geq\; 0,$$

both terms being nonnegative. Hence $\lambda_R(T_2) \geq \lambda_R(T_1)$ for the exact forms.

Two cautions. This does not imply that a *certified numerical* bound improves, since error terms change with the parameters. And it does not order finite matrices of different sizes: changing N changes the compression, so only matrices in the same subspace are comparable by this identity.

6.2 The scalar table

Each R_T entry below is a certified enclosure on a *fixed, frozen* function; no reoptimization is involved, so the columns compare exact forms on the same vector. The Q row comes from the time-domain route and is therefore a **measurement, not a certificate** (Section 4.5); it is shown because the diagnostics below are read off it.

	f_{200}	f_{400}
R_{200}	$1.0310220302459896 \times 10^{-17}$	$1.24275215 \times 10^{-17}$
R_{400}	$2.3520530467310079 \times 10^{-17}$	$1.4330724047355850 \times 10^{-17}$
R_{800}	3.01603×10^{-17}	1.6681×10^{-17}
Q measured	$3.6485917529040770 \times 10^{-17}$	$1.9013829111454644 \times 10^{-17}$

The R_{800} entries carry wider enclosures ($\pm 5.30 \times 10^{-23}$ and $\pm 4.60 \times 10^{-22}$) because the quadrature remainder grows with the cutoff; $R_{200}(f_{400})$ was evaluated as a direct scalar integral on the full degree-438 function, with no truncation and no enlarged matrix. Each column is strictly increasing in T, as Proposition 12 requires.

6.3 What the comparison discards

From (5), with $e_T(f) = q(f) - r_T(f)$ by rigorous interval subtraction. Since q is withdrawn as a certificate (Section 4.5), **every entry in this table is a measurement**, and no statement below rests on one:

	f_{200}	f_{400}
e_{200}	$2.617569722658087 \times 10^{-17}$	6.586308×10^{-18}
e_{400}	$1.296538706173069 \times 10^{-17}$	$4.683105064098794 \times 10^{-18}$
e_{800}	6.3256×10^{-18}	2.333×10^{-18}
e_{400}/e_{200}	$0.49532156 \pm 4.69 \times 10^{-9}$	$0.7110365 \pm 1.61 \times 10^{-8}$
e_{800}/e_{400}	$0.48788 \pm 9.07 \times 10^{-6}$	$0.498 \pm 2.36 \times 10^{-4}$

There is no decay law here. On f_{400} the successive ratios are 0.711 and then 0.498, which is not a constant rate; and the two functions disagree at the *same* doubling $200 \to 400$, giving 0.495 and 0.711, so the ratio is not a function of T alone. That both functions give approximately 0.49 across $400 \to 800$ is two samples of one step, not a rate in T, and comparing differently optimized vectors would establish less still. No uniform operator estimate is inferred from a fixed-vector ratio, and none of these ratios is used in any certified statement.

6.4 What $q_f - r_f$ is, and is not

For f_{400}, $q_f - r_f = 4.6831050640988 \times 10^{-18}$ is the discarded-term identity (5) evaluated on that function, inside its enclosure — a measured quantity, since q is withdrawn (Section 4.5). It is *not* $\lambda_Q - \lambda_R(400)$: both $\lambda_R(400) \le r_f$ and $\lambda_Q \le q_f$ are upper bounds, and two upper bounds do not make their difference an equality between infima.

6.5 Certified ceilings, and the stopping decision

Because f_{400} and f_{200} are admissible, each R_T value in the table is a variational *upper* bound for $\lambda_R(T)$, and any certificate resting only on $Q \ge R_{T,\beta}$ is capped by it. Relative to the lower bound in force:

route	certified ceiling	reference lower bound	headroom
R_{200}	$\lambda_R(200) \le r_{200}(f_{200}) \le 1.0310221 \times 10^{-17}$	1×10^{-17}	3.1023%
R_{400}	$\lambda_R(400) \le r_{400}(f_{400}) \le 1.4330725 \times 10^{-17}$	$1.4330004 \times 10^{-17}$	0.0050%
R_{800}	$\lambda_R(800) \le r_{800}(f_{400}) \le 1.6681461 \times 10^{-17}$	$1.4330004 \times 10^{-17}$	16.409%

Each ceiling above is the *outward-rounded upper endpoint* of the corresponding enclosure, not its midpoint: from $r_{800}(f_{400}) = 1.6681 \times 10^{-17} \pm 4.60 \times 10^{-22}$ the endpoint is $1.6681460 \times 10^{-17}$, quoted outward as $1.6681461 \times 10^{-17}$; quoting the midpoint 1.6681×10^{-17} as a ceiling would be the same error as in Remark 11.

So the $T = 400$ route is within about 0.0050% of exhausted, and the entire $T = 800$ route is capped at a factor 1.1831 over the present lower endpoint; since a reoptimized f_{800} can only sit at or below that ceiling, the realizable gain is at most that. Measured costs on the machine used:

stage	measurement
assembly, $T = 200$, $N = 220$, 800×64 nodes	1338 s, peak 205.5 MiB
assembly, $T = 400$, $N = 440$, same node count	3532 s, peak 393.4 MiB
Rump factorizations, both sectors, $N = 440$	under 50 s
compact $Q(f_{200})$	389 s
cross table (R_{400}, $Q(f_{400})$)	1544 s
scalar leg $R_{200}(f_{400})$, 12800 nodes	933 s
scalar leg $R_{800}(f_{400})$, 51200 nodes	8049 s
scalar leg $R_{800}(f_{200})$, 51200 nodes	3651 s

The assembly grew by a factor 2.64 when N doubled at fixed node count, and the $T = 800$ scalar legs overran the node-count scaling by more than a factor of two, which indicates that per-Bessel cost grows with the argument rather than node count alone. Extrapolating — as an estimate anchored on these measurements, not a guarantee — a full $T = 800$ matrix at $N \approx 880$ with roughly 1600×64 nodes would cost on the order of ten hours of assembly. Against a ceiling of +16.409%, the recommendation recorded in the accompanying repository is to **consolidate at** $T = 400$. Neither the pilot nor any failed certificate here is an intrinsic barrier; this is a cost–benefit judgement from measured data, and a separately scoped $T = 800$ attempt would be a reasonable thing to run with that ceiling stated in advance.

7 The second window

Everything in Sections 3 and 4.1 is run a second time at

$$\widetilde{L} \;=\; \frac{80471895621705018}{10^{17}} \;=\; 0.80471895621705018,$$

a rational chosen strictly below $L_* = (\log 5)/2 = 0.8047189562170501873\ldots$, with $L_* - \widetilde{L} = 7.30037966\ldots \times 10^{-18} < 7.301 \times 10^{-18}$. The choice matters for cost, not for the statement. The prime comb of (2) runs over n with $\log n < 2L$, and $2\widetilde{L} < \log 5$, so $n = 5$ is excluded and

$$A_{\widetilde{L}} \;=\; A_{4/5} \;=\; 2.941973525223620455504,$$

whence the exterior gate of Lemma 8 admits the same cutoff $T = 400$ (the gate threshold is $T_{\mathrm{gate}} = 119.0869\ldots$ at both windows). Had $\widetilde{L}$ been taken above L_*, the comb would acquire $n = 5$, A_L would jump by $2\log 5/\sqrt{5}$ and T_{gate} would rise to $502.39\ldots$, multiplying the assembly cost. At L_* itself the strict inequality still excludes $n = 5$; the jump happens for $L > L_*$.

With $N = 440$, 800 panels of half-length 1/4 and Gauss–Legendre order 64, the certified constants are

$$\beta_{400} = 1.211613695058064549472, \quad \kappa = 7.756529961646111179258\times10^{-29}, \quad d = 1.211613695058064549472,$$

and the two sector shifts, each verified by the preconditioned congruence of Section 3.6 on the assembled block, are

$$a_{\mathrm{even}} = 8.27343868255615234375 \times 10^{-18}, \qquad a_{\mathrm{odd}} = 7.885009765625 \times 10^{-15},$$

with $\|I - G\|_\infty$ equal to 5.0567×10^{-11} and 4.0104×10^{-16} respectively. The strict Schur test then yields Theorem 2(iii). The upper bound (iv) is the closed-form Rayleigh quotient of Section 4.1 at a frozen rational vector in the same 160-mode even cosine basis, at the new window.

Remark 13. $\widetilde{\lambda}^2 = e^{2\widetilde{L}}$ agrees with 5 to about 7.3×10^{-17}; the decimal $\widetilde{\lambda}^2 \approx 4.999999999999999927$ is a rounding of that quantity, not its definition. The definition is the displayed rational $\widetilde{L}$.

8 Conditioned coercivity, by parity sector

The smallness of $\lambda_Q(4/5)$ is carried by a few directions. This is made precise, and certified, by a rank-K device that needs no new assembly.

Lemma 14. *Let A be the compression of R_T to V_N in an orthonormal basis, let $v_1, \dots, v_K$ be any vectors and $c \geq 0$. If $A + c\sum_i v_i v_i^\top \succeq aI$ then $\langle Af, f\rangle \geq a\|f\|^2$ for every f with $\langle v_i, f\rangle = 0$, $i = 1, \dots, K$.*

Proof. On that subspace the rank-K term contributes exactly 0. □

Because the implication holds for *any* v_i, their provenance needs no rigour: only the interval test does. The vectors used here were located by inverse iteration on the midpoint matrix and then frozen as exact rationals with denominator 10^{34}; they are published in the certificates. Combining Lemma 14 with the unchanged complement and coupling constants d and κ of Section 3 through the same strict Schur test gives, at $L = 4/5$:

Theorem 15. *There are explicit rational vectors in the 220-mode even and odd Legendre coordinate bases of the accompanying package such that, for admissible f supported in $[-4/5, 4/5]$ and orthogonal to the listed directions,*

$$\begin{array}{llll} \textit{even sector,} & 4\ \textit{directions:} & Q(f) \geq 0.617553710936882\,\|f\|_2^2 \\ \textit{even sector,} & 5\ \textit{directions:} & Q(f) \geq 0.853492736815556\,\|f\|_2^2 \\ \textit{odd sector,} & 3\ \textit{directions:} & Q(f) \geq 0.106805801391493\,\|f\|_2^2 \\ \textit{odd sector,} & 4\ \textit{directions:} & Q(f) \geq 0.784130096434765\,\|f\|_2^2 \end{array}$$

Over both sectors together, 4 even + 3 odd conditions give $Q(f) \geq 0.106805801391493\,\|f\|_2^2$, and 4 even + 4 odd give $Q(f) \geq 0.617553710936882\,\|f\|_2^2$.

Short forms must round *down*: 0.6175 is a valid abbreviation of the four-condition even constant and 0.6176 is not.

Remark 16. These are per-sector statements, and four conditions on the combined space cannot give the even constant: by min–max the odd sector carries minuscule eigenvalues of its own, so the combined-space bound with 7 conditions is the smaller of the two sector constants.

Remark 17. Removing directions changes the minimum over what remains; it does *not* raise $\lambda_Q(4/5)$, which stays inside the bracket of Theorem 2. And the conditioning spaces here are the spans of explicitly listed numerical vectors. They are *not* identified with the prolate conditioning spaces of [1, §5.3]; that identification would need a separate proof or a separate certificate, and neither is offered here.

9 An eigenvalue count at $L = 4/5$

Theorem 15 bounds a count from above. By Courant–Fischer, four conditions leaving the infimum at least 0.6175 means at most four even eigenvalues of Q lie below 0.6175, in particular at most four below 10^{-2}; the three-condition odd constant gives at most three odd. For the other direction one needs a test *space*, and the closed-form evaluator supplies one.

Proposition 18. *Let M_N be the matrix of Q on the first N even cosine modes in an orthonormal basis, computed in closed form as in Section 4.1. Since M_N is a compression of Q, min–max gives $\lambda_j(M_N) \geq \varepsilon_{\text{even}}(j)$ for every j. Hence for any K-dimensional $W \subset \mathbb{R}^N$,*

$$\varepsilon_{\text{even}}(K) \;\leq\; \lambda_K(M_N) \;\leq\; \lambda_{\max}(M_N|_W),$$

and $\lambda_{\max}(M_N|_W) \leq c$ is certified by the $K \times K$ test $cG - B \succ 0$ with $G = W^\top W$ and $B = W^\top M_N W$.

At $N = 160$, working precision 2920 bits, with W a frozen rational 4-dimensional space (denominator 10^{30}; the Gram matrix G is verified positive definite, so $\dim W = 4$), the test certifies

$$\varepsilon_{\mathrm{even}}(4) \;\le\; 0.00311839891820 \;<\; 0.003119 \;<\; 10^{-2}.$$

Together with the four-condition bound:

Theorem 19. *At $L = 4/5$, exactly four eigenvalues of Q in the even sector lie below 10^{-2}.*

In the odd sector only **at most three** is certified: the closed-form evaluator implements the even cosine basis only, so no odd compression of Q is available here and the matching lower side is absent. Counts observed in the compression of the comparison form R_T are *numerical*: since $Q \ge R_T$, the eigenvalues of a compression of R_T bound those of Q from neither side for $j \ge 2$. Neither count is transferred to $\widetilde{L}$ or to L_*; that would need its own certificate.

10 Normalization dictionary, and relation to Andrews–Connes–La Gatta

[1] studies the Weil quadratic form on the multiplicative window $[\lambda^{-1}, \lambda] \subset \mathbb{R}_+^*$ with $d^*u = du/u$, in the normalization $\Psi = W_{0,2} - W_{\mathbb{R}} - \sum_p W_p$ of their equations (6)–(8). Writing $t = \log u$ and $R(t) = f(t)$, that form and the form (1) of Section 2 are the same object:

Proposition 20 (Dictionary)**.** *Let $g(u) = f(\log u)$ and $\lambda = e^L$. Then $\|g\|^2_{L^2(d^*u)} = \|f\|^2_{L^2(dx)}$ and $QW_\lambda(g,g) = Q(f)$, with the two pole signs matching sector by sector.*

Proof. In the additive variable, $F(e^s) = C(s) = \int R(t)R(t+s)\,dt$. Their $W_{0,2}(F) = 2\big(\int Re^{u/2}\big)\big(\int Re^{-u/2}\big)$, which is the pole term (3) with $s = +1$ for even R and $s = -1$ for odd R, since then $\int Re^{u/2} = -\int Re^{-u/2}$. Their $W_p(F)$ is term by term the prime sum of (2). For the archimedean part, their Lemma 8.13 writes $-W_{\mathbb{R}}$ as the Fourier multiplier $m(\tau) - c_0$ with $m(\tau) = \int_0^\infty (1 - \cos\tau s)\, e^{s/2}/\sinh s\; ds$ and $c_0 = \log 4\pi + \gamma_E + \int_0^\infty (e^{s/2} - 1)/\sinh s\; ds$, so

$$m(\tau) - c_0 \;=\; \int_0^\infty \frac{1 - e^{s/2}\cos\tau s}{\sinh s}\, ds \;-\; \log 4\pi \;-\; \gamma_E.$$

From $\psi(z) = -\gamma + \int_0^\infty (e^{-u} - e^{-zu})/(1 - e^{-u})\, du$ at $z = \frac{1}{4} + \frac{i\tau}{2}$, substituting $u = 2s$ and using $1 - e^{-2s} = 2e^{-s}\sinh s$,

$$\Re\,\psi\Big(\tfrac{1}{4} + \tfrac{i\tau}{2}\Big) \;=\; -\gamma + \int_0^\infty \frac{e^{-s} - e^{s/2}\cos\tau s}{\sinh s}\, ds.$$

The difference of the two displays is $\int_0^\infty (1 - e^{-s})/\sinh s\; ds - \log 4$, and $(1 - e^{-s})/\sinh s = 2e^{-s}/(1 + e^{-s})$ integrates to $2\log 2 = \log 4$. Hence $m(\tau) - c_0 = \Re\,\psi(\frac{1}{4} + \frac{i\tau}{2}) - \log\pi$, which is the multiplier Ψ_L of (2) up to the prime comb already matched. □

Proposition 20 is an *analytic identity*. It was additionally cross-checked numerically: an independent implementation of (6)–(8), sharing no code with this package, reproduces the certified values of Q at the Legendre test functions φ_0, φ_2 and $\varphi_0 - \varphi_2$ to about thirty digits, matching the stored enclosures to their serialized width. The cross-check corroborates the identity; it does not replace the proof.

Under the dictionary our window $L = 4/5$ is $\lambda^2 = 4.95303242439511480365$ and $\widetilde{L}$ is $\lambda^2 = 5 - 7.3 \times 10^{-17}$, just below the smallest window tabulated in [1].

What is comparable, and what is not. The eigenvalues reported in [1, §4] are eigenvalues of finite log-Fourier compressions; by min–max they bound the limiting eigenvalues from above only, as those authors state. Their value at $\lambda^2 = 5$, recovered from a printed $-\log\varepsilon$, is $\approx 9.1663 \times 10^{-18}$; it lies inside the bracket of Theorem 2(iii)–(iv) at our nearby window. That placement is **numerical consistency** between a reconstructed finite-compression estimate and a certified bracket at a slightly different parameter — not a certified comparison at identical parameters, and not an independently known value of the infinite-dimensional infimum.

Their Hypothesis 3.4 compares $\varepsilon_\lambda(j)$ with the uncertainty-operator rungs $\delta_{4j}(\lambda)$ up to constants and a factor $\log\lambda$. It does *not* assert that the ratio of two Weil minima equals the ratio of the corresponding rungs; that would need the proportionality factor to be the same at both windows, which is a strictly stronger approximation. In particular the rung ratio $\delta_4/\delta_4 = 0.579867\ldots$ between our two windows is a **numerical rung-ratio reference**, not a prediction, and it happens to lie inside the interval of Corollary 6. All comparisons against δ_{4j} in the accompanying report are labelled numerical, because the rung values there are high-precision rather than interval enclosures.

11 Reproducibility and limitations

11.1 Artifacts and verification

Every decisive quantity in this note is published as a midpoint–radius enclosure in machine-readable form, together with the exact rational input vectors and the code that produced them. Printed decimals in this note are summaries; the enclosures are authoritative.

Two layers of checking ship with the package. The *repository suite* — eight scripts, 176 checks — recomputes the phase-1 to phase-6 chain from the frozen artifacts and exits nonzero on any failure. On top of it the *release gate* `phase7/acceptance.py` re-verifies every claim of this note from the packaged files rather than from values held in memory: the repository suite itself; the closed-form upper bound re-certified from its stored exact rationals; a Rump replay of *both* improved unconditioned $L = 4/5$ shifts, which the phase-4 verifier does not cover because it tests the earlier committed shifts and because the strict Schur test assumes $A \succeq aI$ rather than checking it; all eight Schur checks from their own stored a, d, κ, μ, with rejection required at $\mu = a$; the four conditioned rank-K bumps rebuilt from the stored rational vectors; both second-window blocks; the second-window upper quotient; the count certificate with its Gram matrix; the directed-decimal checks and the ratio of Corollary 6; and the provenance and remainder fields the chain depends on. On the run accompanying this version the gate reports 61 **checks,** 0 **failed, in** 207 **seconds**, and the repository suite inside it reports 176 passing checks.

All of this is *frozen-artifact replay*, not a cold rebuild: the three matrix assemblies are not re-run (measured at 3532 s for $T = 400$ at $L = 4/5$, 3678 s at $\widetilde{L}$, 1343 s for $T = 200$), and the scripts that regenerate them are named in the package instructions. The verifiers are not independent implementations: they share the assembly and constants modules with the computation they check, and no independence is claimed for them. One script re-derives every decimal quoted above from the artifacts and additionally checks each quoted bound's *rounding direction* and its scope and parity label; it is what caught the mis-quoted upper endpoint of Appendix A, item 6.

Within those scripts two kinds of check must be distinguished. Most recompute a *certificate-producing inequality* — a Schur sign, a Cholesky residual, a tail bound — in ball arithmetic, and a failure there invalidates a claim. A few are *auxiliary*: they compare the Arb assembly against the repository's independent `mpmath` implementation, or assert that a diagnostic spectrum is finite. Those use floating point and settle nothing about the certificates; one of them formerly compared two matrices after converting both to ordinary doubles while asserting agreement below 10^{-30}, a threshold that no double-precision comparison can reach. It now compares at

full precision and states the tolerance its inputs actually support (Appendix A, item 8).

The verifiers fail closed. Any NaN, infinite radius, indeterminate enclosure, or unresolved sign produces an explicit failure or a refinement request; no skipped cell, panel or pivot is interpreted as success. This matters because an earlier version of the computation silently skipped cells where a library digamma call returned NaN on wide balls and thereby produced a wrong constant that was *used* (Appendix A, item 2).

11.2 Environment

Linux, Python 3.14.7, `mpmath 1.3.0`, `python-flint 0.9.0` (FLINT/Arb). Floating-point `mpmath` is used to *locate* candidate vectors, which are then frozen as exact rationals and certified in Arb, and in the auxiliary cross-implementation checks described above. It enters no certificate-producing inequality; it does appear in a small number of auxiliary verifier assertions, which is a weaker statement than the one made in earlier versions of this note. Arb working precisions on sign-critical paths range from 256 to 2000 bits and are stated per stage in the artifacts.

11.3 Limitations

1. **Two windows.** Everything is at $L = 4/5$ and $L = \widetilde{L}$. There is no propagation theorem and nothing follows at any window above $\widetilde{L}$ except the upper bound of Proposition 5. In particular *no positive lower bound at* $L_* = (\log 5)/2$ follows from these certificates.

2. **An excluded route, still open.** The time-domain evaluator of Section 4.2 omits the truncation error of its outer panelled rule, so every value it produced remains uncertified (Section 4.5) and is excluded from every claim above. A rigorous outer remainder is still open; it is off the critical path, since closing it would restore a second route to a bracket this note already has.

3. **No odd count from below.** Theorem 19 is even-sector only. The closed-form evaluator implements the even cosine basis, so there is no odd compression of Q here and only "at most three" is certified in the odd sector.

4. **The conditioning spaces are not identified with prolate spaces.** See Remark 17. The constants of Theorem 15 hold for the explicitly listed rational vectors; no identification with [1, §5.1–5.3] is proved or certified.

5. **Comparisons against the uncertainty-operator rungs are numerical.** The rung values $\delta_{4j}(\lambda)$ used in the accompanying report are high-precision, validated against the printed values of [1], but are *not* interval enclosures. Every ratio against them is labelled numerical, and none enters a certified claim.

6. **Rational windows only.** The assembly is parameterized by a rational L; $L_* = (\log 5)/2$ is irrational and cannot be used directly, which is why the second window is the rational $\widetilde{L}$ just below it.

7. **An imported normalization.** The explicit-formula pairing in the convention of Section 2 is imported from the accompanying repository, not rederived there. It is now additionally supported by Proposition 20, which identifies it with an independently published normalization and is corroborated numerically to about thirty digits; that is evidence for the convention, not a derivation of it.

8. **An unverified external matrix step.** Zhu describes supplementary source code and certificates in [10, §17]; these were not obtained or executed here, so the matrix step of [10, Thm. 1.1] is not reproduced. The present certificates are independent of it.

9. **Shared arithmetic.** The verifiers share library code with the computation they check, and are reported as replays, not as independent implementations.

10. **No arithmetic content.** The only analytic sharpening proved here is the monotone gate of Lemma 8, which moves a constant by under one percent and does not touch the doubly exponential growth of $T_{\mathrm{gate}}(L) = 2\pi e^{A_L}(1 + o(1))$ that governs the cost in L. That growth, and not the finite block, is what caps this method.

11. **Scope.** No zero-free region, no verified range of zeros, no statement about the Riemann hypothesis, and no claim to have settled any open question of [1].

11.4 Revision history

Versions 1.0.x through 1.2.0 were internal drafts, never deposited. Version 2.0.0 is the first archived release and the first with a different scope: two windows rather than one. The stages are kept distinguishable on purpose — the versions that carried the *original omission*, the version that made the *interim correction*, and the release that *excludes* the affected route. No version is rewritten in place.

status Versions 1.0.x–1.2.0 are **internal drafts**; none was deposited, released or archived. Version 2.0.0 is the first archived release.

1.0.x carried the omission. The upper bound 1.9014×10^{-17} and everything derived from it rested on an incomplete error budget. Those versions should not be cited for the upper bound.

1.1.0 interim correction. The affected claims were withdrawn and the endpoint reverted to a quadrature-free route.

1.2.0 improved, repair still open. The quadrature-free endpoint was tightened past the withdrawn value. The outer quadrature remainder itself is *not* repaired; that would be a later version.

1.0.0 (2026-11-10) First version.

1.0.1 (2026-11-10) Citation and provenance revision only. Adds the two earlier technical notes whose implementation and parity conventions this computation inherits ([11] and [12]), the lineage paragraph in Section 1.3, and the corresponding citations in Sections 2 and 4. No numerical result, bound, artifact or limitation changed; the certified bracket and the parity statement are exactly those of version 1.0.0.

1.0.2 (2026-11-10) Corrections after reading [10] in full, which had previously been consulted only through its abstract and rendered HTML. Three attributions were too weak and are strengthened: [10, Thm. 6.2] certifies the parity *separation* at this very window, with a odd-sector constant, and establishes simplicity and evenness of the ground state (Sections 1.3 and 5); [10, §6.2] already evaluates its variational upper bounds by the time-domain route used in Section 4; and the supplementary materials of [10, §17] are described there, rather than absent. Section 1.4 is new and compares the two sets of bounds at this window. Again no numerical result, bound or artifact changed.

1.1.0 (2026-11-10) **The certified upper endpoint changes.** The time-domain evaluator was found to omit the truncation error of its outer quadrature (Section 4.5), so every value it produced is withdrawn as a certificate and the upper bound reverts to the closed-form witness of Section 4.1: 2.1130×10^{-17} in place of 1.9014×10^{-17}. The bracket widens from 4.914×10^{-18} to 7.03×10^{-18}, the parity separation is re-established on the closed-form

witness with margin 1.87×10^{-18}, and Section 1.4 is updated. The lower bound, the odd-sector bound, the R_T values and the ceilings are unaffected. Four further corrections: the Hermitian pole term for complex f (Section 2), the degree of the autocorrelation, outward rounding of the ceilings, and the scope of the verification claims (Section 11).

1.2.0 (2026-11-10) **The certified upper endpoint improves past the withdrawn one.** Rather than repair the unbounded quadrature remainder, the endpoint is tightened along the route that has no quadrature to bound: the closed-form evaluation is pushed from 28 even cosine modes to 160, giving $U = 1.6731255341 \times 10^{-17}$ in place of 2.1130×10^{-17}, and in place of the withdrawn time-domain value 1.9014×10^{-17} as well. The bracket narrows to 2.64×10^{-18}, 80.9% inside the published one, and the parity margin grows to 6.27×10^{-18}. The lower bound, the odd-sector bound, the R_T values and the ceilings are unchanged. The outer quadrature remainder of Section 4.5 remains open; it is no longer on the critical path.

2.0.0 (2026-11-12) **First archived release; the scope becomes two windows.** New: the second window $\widetilde{L}$ with its own two-sided bracket (Section 7); the ratio Corollary 6; the monotonicity and coverage statement (Proposition 5); per-sector conditioned coercivity (Section 8); the even-sector count (Section 9); and the proved normalization dictionary to [1] (Section 10).

Sharpened, with no reassembly — the frozen $T = 400$ blocks admitted larger Rump shifts than had been extracted: the $L = 4/5$ lower endpoint from 1.41×10^{-17} to $1.43300048827856 \times 10^{-17}$, and the odd-sector constant from 2.3×10^{-17} to $1.38569999999861 \times 10^{-14}$, which now exceeds [10, Thm. 6.2]'s odd constant by a factor 1.68. The $T = 400$ even route is thereby exhausted: the certified floor and the independently measured ceiling $r_{400}(f_{400})$ agree to a relative 5.0×10^{-5}, so the $T = 800$ headroom is +16.4%, not +18.3%.

Corrected in the exposition: published μ values are now short exact decimals with explicit Schur margins. A μ bisected to the supremum has margin $a - \mu^* = \kappa^2/(d - \mu^*)$, which is $\approx \kappa^2/d$ and so $\sim 10^{-41}$ of a; that cannot survive serialization, and the earlier export could not be replayed. Every printed bound is now directed, replacing in particular an invalid short upper form 1.6731×10^{-17} by 1.674×10^{-17}; the eigenvalue counts are stated as "exactly four even" (both sides certified) and "at most three odd" rather than as counts read off a compression of R_T; and the rung-ratio value 0.579867 is labelled a numerical reference rather than a prediction of [1, Hyp. 3.4]. The front-page draft notice is replaced by a scope notice; the time-domain route is excluded rather than carried. Appendix A records the new failure modes.

A Error history

Errors found during this work are recorded because several of them produced plausible wrong numbers rather than failures, and because the countermeasures are part of what the artifacts are for. Items 1–5 are library- or discipline-level defects; items 6–10 are corrections to claims; items 11–13 were found while preparing version 2.0.0. A longer ledger, including retracted statements $R1$–$R13$, is kept with the artifacts.

1. **Pole-coordinate phase.** The form is assembled in the coordinates $y_n = \sigma_n x_n$ with $\sigma_n = (-1)^{\lfloor n/2 \rfloor}$, so the pole vector requires the same phase. Omitting it is invisible at degree 0 and wrong from degree 2 onward. It produced a genuinely negative pivot (-0.3949) that would have been reported as indefiniteness. Caught by comparing $|F(t)|^2$ and $F(i/2)$ against direct integration; the degree-2 checks of Section 4 now cover it.

2. **Digamma on wide balls.** The library digamma returns NaN on balls whose real part has radius $\gtrsim 0.1$, even strictly inside the right half-plane. A comparison loop silently skipped those cells and returned $M \approx 1831$ instead of 8.83 — and the wrong value was used. Replaced by the explicit recurrence $\psi(z) = \psi(z+K) - \sum_{k<K}(z+k)^{-1}$ with adaptive K, and by a covering-free monotonicity bound for M.

3. **Bessel functions on wide balls.** Interval evaluation of j_n on the complex ellipses is useless here (width $\sim 10^{32}$ at $|w| \sim 80$, unchanged at 3000 bits). Replaced by the uniform Cauchy–Schwarz bound (7).

4. **Interval factorization.** Naive interval LDL^{T} is unusable at these dimensions (radii amplify about 6.5× per pivot, reaching $O(1)$ by step 47). Replaced by the Rump preconditioned congruence.

5. **Precision ownership.** The Arb context precision is a process global. Routines set it on entry and did not restore it, so a callee silently changed its caller's context; this produced four separate wrong constants in this project, each plausible-looking rather than an error (a complement constant at $[\pm 7.4 \times 10^{132}]$, an even-sector d at $[\pm 2.8 \times 10^{183}]$, and two coefficient sets built at 53 bits). The repair is an owned-scope wrapper that asserts restoration on normal return *and* on exception, caches keyed by precision so a low-precision value can never be served to a high-precision caller, and an explicit *unresolved* result for any wide or non-finite enclosure. The lesson, and the harder half, is that *an input constructed outside the scope carries the ambient precision's radius, and entering a higher-precision scope afterwards does not recover it*: own your inputs as well as your scope.

6. **The upper endpoint, twice mis-quoted.** Two distinct errors, both in the safe direction numerically but both wrong as stated. (a) Internal drafts quoted the enclosure *midpoint* $1.90138291114546440 \times 10^{-17}$ as the bracket's upper endpoint; a midpoint is not an upper bound for its own ball. (b) A separate record quoted $1.9013829111454646892 \times 10^{-17}$ as "the exact upper endpoint"; that value came from re-parsing the serialized ball at the ambient 53 bits, which rounds the midpoint and then inflates the endpoint by about 2.8×10^{-33} — the precision-ownership defect of item 5 recurring in the very routine that documents it. It recurred twice more while that was being repaired: once extracting an endpoint after an owned scope had exited, once parsing a stored ball string at the ambient precision. Four occurrences, which is why deserialisation and endpoint extraction now have their own entry points in the precision module instead of being written inline. See Remark 11.

7. **An omitted quadrature remainder.** The time-domain evaluator adds an explicit bound for the integrable head on $[0, \eta]$ but none for the truncation error of its outer panelled Gauss–Legendre rule, while its module docstring asserted that a Bernstein-ellipse remainder was used. Ball arithmetic does not supply that term: it encloses the nodes, weights and rounding, not the difference between the rule and the integral. The published radii were therefore incomplete, and the bound they supported is withdrawn (Section 4.5). The certified endpoint now comes from a route that performs no quadrature. This is the one item in this list that changed a headline number.

8. **A float comparison asserting an unreachable tolerance.** An auxiliary verifier check compared the Arb assembly with the repository's `mpmath` assembly by converting both to ordinary doubles and requiring agreement below 10^{-30}. Doubles carry about sixteen digits, so the comparison could only report exact equality or a difference near 10^{-16}; the threshold was unreachable and the check was weaker than it read. It now compares at full precision against the tolerance its inputs support.

9. **A retracted sparsity target.** An earlier proposed next step — a power-saving sparsity bound on the near-resonant set of the prime comb — is false, and is retracted in full: by continuous Kronecker–Weyl equidistribution that set has positive density, confirmed numerically here to four significant figures across four decades of T. The replacement is the empty-exterior gate of Section 3, which needs no such estimate.

10. **Overstated interpretations, corrected.** Four further claims made alongside earlier versions were withdrawn or corrected: that no bound on the measure of a bad set could control $\int_E |F|^2$ (false — standard finite-measure and local-occupation estimates do exactly that); that a Schur *threshold* was a lower bound (it is a break-even point, and the final test needs strict inequalities, hence (6)); that lowering β would buy a larger a (refuted by Proposition 9 — the $\beta\|f\|_2^2$ term had been ignored); and an accusation that the library's adaptive integrator returned a silently wrong value (withdrawn — it could not be reproduced; the wrong numbers came from a run that still carried the pole-phase error of item 1, at a different cutoff, with output from two runs conflated).

11. **Precision ownership, in new drivers (version 2.0.0).** The defect of item 5 recurred four times in driver code written outside the repaired `constants()` path, each time producing a silently vacuous or unreplayable record rather than an error. (a) Certificate constants recomputed inline, before any routine had set the working precision, gave d a radius of $3 \times 10^{+230}$ and $\kappa = \mathrm{NaN}$ in one sector only — call-order dependent — so the Schur step could not close and the sector reported a vacuous $\mu \geq 0$. (b) A one-sided shift search that doubled upward and stopped at the first failure certified *nothing* when the start guess was 1.4% above the true $\lambda_{\min}$. (c) Serialized matrix entries re-parsed after the owned scope had exited carried the ambient 53-bit radius: entry radii inflated from 5.12×10^{-49} to 7.18×10^{-19}, a factor 1.4×10^{30}, and no shift could be certified at all. (d) A μ bisected to the Schur supremum has margin $a - \mu^* = \kappa^2/(d - \mu^*)$, which is $\approx \kappa^2/d \approx 10^{-58}$ here, i.e. $\sim 10^{-41}$ of a; serialized at 25 digits its truncation radius $\sim 3 \times 10^{-46}$ is 10^{12} times the margin, so the published μ was a ball straddling a and the strict test $a > \mu$ *failed* on replay.

    All four were failures *to certify*, not false certifications: the positive-definiteness routine sets its own precision, so every shift it reported as passing was valid, and in (d) the underlying certification had been performed correctly at full precision while only its serialization was unusable. The countermeasures are now rules: every driver takes its constants through `constants()` under owned precision with tightness assertions; every re-parse of a serialized ball names its precision; shift searches are two-sided; and a published constant must carry a margin large enough to survive serialization, which is why the μ values of this version are short exact decimals at $a(1 - 10^{-12})$ rather than at the supremum. This is the only blanket statement made here: it covers the four items above and is not a guarantee about defects not yet found.

12. **A downward-rounded upper bound (version 2.0.0).** The short form 1.6731×10^{-17} was printed in several places as an upper bound for $U_0 = 1.67312553407068\ldots \times 10^{-17}$, which it is not: a decimal rounded *down* cannot bound a quantity from above. This is item 6 recurring. Corrected to 1.674×10^{-17}, the value the artifact already carried, and the release gate now checks the direction of every short bound and explicitly rejects both 1.6731×10^{-17} as an upper form and 0.6176 as a short lower form for 0.617553710936882.

13. **Counts asserted from the wrong operator (version 2.0.0).** The statement that exactly four even and exactly three odd eigenvalues lie below 10^{-2} was first made from the eigenvalues of a compression of the comparison form R_T. Since $Q \geq R_T$, those bound the eigenvalues of Q from neither side for $j \geq 2$, so the claim was unsupported. The even

count is now certified on both sides (Theorem 19); in the odd sector only "at most three" is claimed, and the compression counts are labelled numerical.

References

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